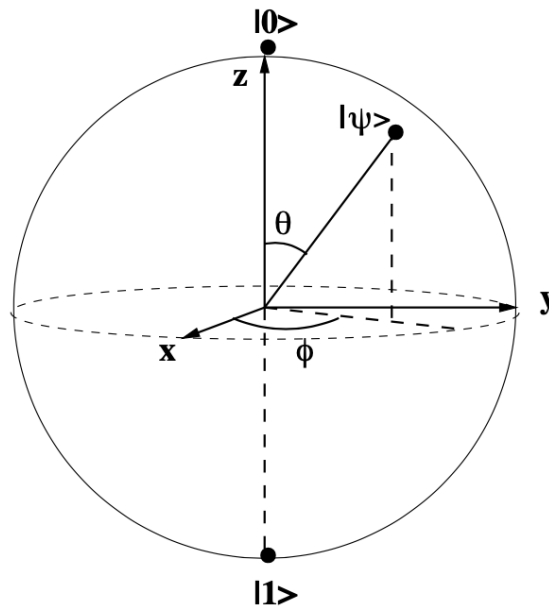


Problem 1.

■ Show that there is a one to one correspondence between qubits

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$$

and the points on the unit sphere in $\mathbb{R}^3$, called the *Bloch sphere*, with θ and ϕ as the spherical coordinates of a point of the sphere; see the figure below.



■ Show that the matrix representation of the NOT gate in the basis $|0\rangle, |1\rangle$ is given by $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. What is its matrix representation in the basis $|+\rangle, |-\rangle$?

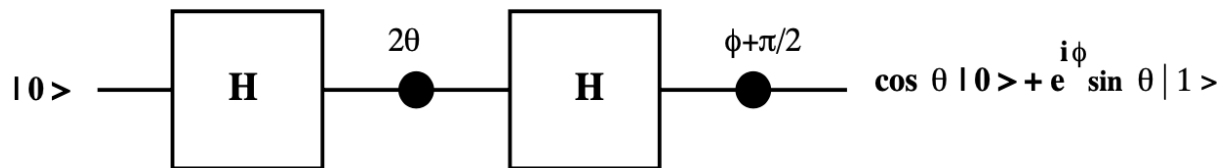
Problem 2.

■ Show that any unitary matrix U can be expressed as

$$U = \begin{pmatrix} e^{i(\alpha-\beta/2-\delta/2)} \cos \gamma/2 & -e^{i(\alpha-\beta/2+\delta/2)} \sin \gamma/2 \\ e^{i(\alpha+\beta/2-\delta/2)} \sin \gamma/2 & e^{i(\alpha+\beta/2+\delta/2)} \cos \gamma/2 \end{pmatrix}$$

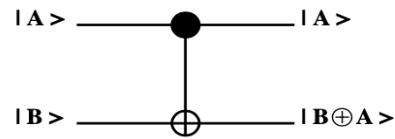
where α , β , δ and γ are real numbers.

■ Verify the output, up to a global phase, of the following quantum network.



Problem 3.

■ Check that the matrix representing CNOT in the standard computational basis is the one given in Figure 1 and that it is unitary.



$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Figure 1: The CNOT gate

■ Show that the circuit on the left in Figure 2, swaps the two qubits. It is denoted by the figure on the right. What is its matrix representation?

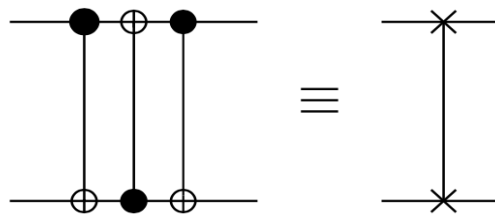


Figure 2: Swapping Gate

Problem 4. Measurement Postulate

a) Consider the two qubit register

$$|\psi\rangle = \frac{1}{3}|00\rangle + \frac{2}{3}|10\rangle - \frac{2}{3}|11\rangle.$$

- (i) Calculate the probability of measuring 1 on the first qubit and establish the state of the register after this measurement.
- (ii) Now measure the second qubit. What is the probability of obtaining results 0? What is the final state of the register?

b) Design a circuit which realizes the teleportation of a qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ from A to B using the entangled pair $|\beta_{11}\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$.

c) Find out which of the following two sets is a computational base

$$\left\{ \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle), \frac{1}{\sqrt{2}}(i|0\rangle - |1\rangle) \right\},$$

$$\left\{ \frac{1}{2}(|0\rangle + \sqrt{3}|1\rangle), \frac{1}{2}(\sqrt{3}|0\rangle - |1\rangle) \right\}$$

Calculate the results of measuring $\alpha|0\rangle + \beta|1\rangle$ in that base and their probabilities.

d) Consider a two qubit register in the state

$$|\phi\rangle = \frac{1}{\sqrt{12}}(3|00\rangle + |01\rangle + |10\rangle - |11\rangle).$$

Suppose that Alice and Bob can operate on the first and second qubit, respectively.

- (i) What is the probability that both Alice and Bob measure 1 on their own qubit?
- (ii) If Alice and Bob apply **H** on their qubit before measuring it, what is the probability that both achieve the result 1?